(15) Energy & Momentum in EM Waves

We have learnt:

$$W_E = \frac{1}{2}\underline{D}\underline{E} = \frac{1}{2}\varepsilon E^2$$

in linear medium

$$W_m = \frac{1}{2}\underline{B}\underline{H} = \frac{1}{2\mu}B^2$$

So the EM wave carries Energy!

Which one is higher W_E or W_M?

$$\underline{B} = \frac{\underline{k}\times\underline{E}}{\omega} \rightarrow |B| = \frac{k\cdot E}{\omega} = \frac{E}{v_f} = \sqrt{\varepsilon\mu}\,E$$

The magnetic en. density:

$$W_M = \frac{1}{2\mu}B^2 = \frac{1}{2\mu}\left(\sqrt{\varepsilon\mu}E\right)^2 = \frac{1}{2\mu}\varepsilon\mu E^2 = \frac{1}{2}\varepsilon E^2 = W_E$$

Not only the average en. density is equal but also the space and time dependent energy is instantaneously equal!

The total energy density is the sum:

$$W_{EM} = \frac{1}{2}\varepsilon E^2 + \frac{1}{2\mu}B^2 = \varepsilon E^2 = \varepsilon E\cdot v_f B$$

It is reasonable to suppose that this energy is propagating in a direction $\parallel \underline{k}$, at v_f.

as $\underline{k} \parallel \underline{E} \times \underline{B}$ in hom isotrop lin. mat.

we can define the energy-flow-density vector, or Poynting vector as:

prop. direction $\parallel \underline{E} \times \underline{B}$

amplitude: energy transfer rate for unit surface:

$$\underline{S} = \varepsilon \cdot v_f^2 \, \underline{E} \times \underline{B}$$

$$\underline{S} = \frac{\varepsilon}{\varepsilon\mu} \underline{E} \times \underline{B} = \underline{E} \times \underline{H}$$

$[S] = \frac{W}{m^2}$ ¹ $v_f \cdot w_{EM}$

Its physical meaning:

$$\frac{dE_{EM}}{dt} = \iint \underline{S} \, d\underline{s}$$

$\frac{dE_{EM}}{dt}$

$$\left(\frac{dE_{EM}}{\partial t}\right)_{enclosed} = -\oint\!\!\oint \underline{S} \, d\underline{s}$$

Continuity eqn. Conservation of en.

Is it embedded in Maxwell's equations?

AM $\quad \overset{curl}{\cancel{rot}}\, \underline{H} = \frac{\partial \underline{D}}{\partial t} + \underline{j} \;/ \cdot \underline{E}$

F: $\quad \overset{curl}{\cancel{rot}}\, \underline{E} = -\frac{\partial \underline{B}}{\partial t} \;/ \cdot \underline{H}$

$\underline{E} \cdot \overset{curl}{\cancel{rot}}\, \underline{H} = \underline{E}\dot{\underline{D}} + j\underline{E}$

$\underline{H} \cdot curl\, \underline{E} = -\underline{H}\dot{\underline{B}}$

$$\underbrace{(\underline{E}\, curl\, \underline{H} - \underline{H}\, curl\, \underline{E})}_{-div(\underline{E} \times \underline{H})} - \underline{E}\dot{\underline{D}} - \underline{H}\dot{\underline{B}} = \underline{j}\underline{E}$$

hom. lin ~~diel~~ material

$$\frac{\partial}{\partial t}(W_e + W_M) = \varepsilon \underline{E}\dot{\underline{E}} + \frac{1}{\mu}\underline{B}\dot{\underline{B}}$$

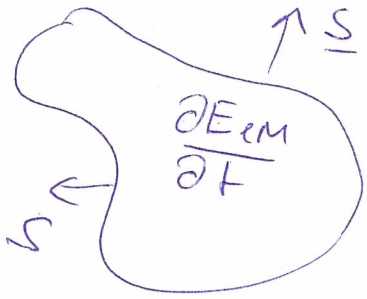
$$\frac{\partial}{\partial t} W_{EM} = -\operatorname{div} \underline{E} \times \underline{H} - \underline{j}\underline{E}$$

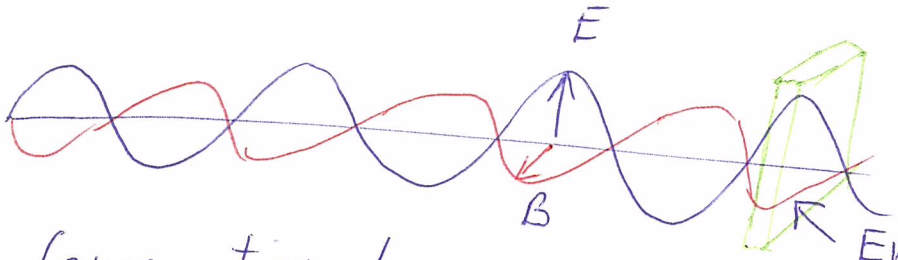
} cont. eq. for energy

change rate of energy density — power density flowing out — Joule heat density

$$\iiint \frac{\partial}{\partial t} W_{EM}\, dV = -\iiint \operatorname{div} \underline{E} \times \underline{H}\, dV - \iiint \underline{j}\underline{E}\, dV$$

$$\frac{\partial E_{em}}{\partial t} = -\oint\!\oint \underline{E} \times \underline{H}\, d\underline{s} - \frac{\partial E_{joule}}{\partial t}$$





Energy is not const!

Conservation of energy: only for closed systems!

$\frac{\partial E_{em}}{\partial t} \sim \cos^2$

$\underline{E}$ and $\underline{B}$ reaches max simult.

What we can measure:

Time average of E_{em} and $\underline{S}$

$$\underline{S} = \underline{E} \times \underline{H} = \underline{E}_0 \times \underline{H}_0 \cdot \cos^2(\underline{k}\underline{r} - \omega t) = \underline{E}_0 \times \underline{H}_0 \cdot \frac{1}{2}(1 + \cos 2(\ldots))$$

$$\# \underline{E}_0 \times \underline{H}_0 \cdot \frac{1}{2} + \underline{E}_0 \times \underline{H}_0 \cdot \frac{1}{2} \cdot \cos(2\underline{k}\underline{r} - 2\omega t)$$

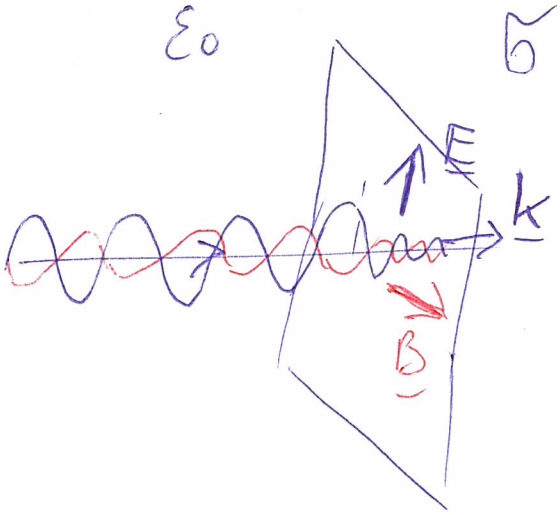
const energy flow

energy flowing back and forth at 2ω

$$\langle \underline{S} \rangle = \frac{1}{2} \underline{E}_0 \times \underline{H}_0$$

Momentum of EMW

Consider an EMW incident on a conducting surface from vacuum!



We know that the wave will be absorbed and its energy dissipated within a thin layer at the surface ($\sim \delta_{skin}$)
(Suppose there is no reflection)
Is there any force or pressure?

Let's investigate the motion of an e^- close to the surface!

The electric force acting on an e^-:

$$\underline{F}_e = q\underline{E} = q E_0 \cdot \sin \omega t \quad (\text{at } x = 0)$$

Ohm: $\underline{j} = \sigma \underline{E}$ $\qquad$ $\underline{j} = n v_d q$, $\sigma = \frac{q^2 n}{k}$ (drag modell (Drude))

$$\Downarrow \qquad \Leftarrow$$

$$n \underline{v}_d q = \sigma \underline{E} = \frac{q^2 n}{k} \underline{E}$$

$$\underline{v}_d = \frac{q}{k} \underline{E} = \frac{q}{k} \cdot \underline{E}_0 \sin \omega t$$

Magnetic (Lorentz) force acting on the e^-:

$$\underline{F}_B = q \cdot \underline{v}_d \times \underline{B} = q \frac{q}{k} \underline{E}_0 \sin \omega t \times \underline{B}_0 \sin \omega t =$$

$$= \frac{q^2}{k} \underline{E}_0 \times \underline{B}_0 \cdot \sin^2(\omega t) \qquad \underline{E}_0 \times \underline{B}_0 \parallel \underline{k} \; !!!$$

An oscillating force pointing towards the inwards metallic surface!

$$|\underline{F_B}| = \frac{q^2}{k}\frac{E_0^2}{v_{ph}}\cdot\sin^2(\omega t)$$

$v_{ph} \to c$

$$a\times(b\times c) = (a\cdot c)b - (ab)c$$

$$\underline{F_B} = \frac{q^2}{k}\underline{E_0}\times\underline{B_0}\cdot\sin^2(\omega t) = \frac{q^2}{k}\underline{E_0}\times\left(\frac{\underline{k}\times\underline{E_0}}{\omega}\right)\cdot\sin^2(\omega t)$$

$$= \frac{q^2}{k}E_0^2\cdot\underline{k}\,\sin^2\omega t$$

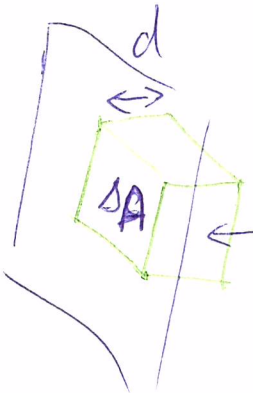
$$\underline{F_B} = q\,\underline{v_d}\times\underline{B} = q\,\underline{v_d}\times\left(\frac{\underline{k}\times\underline{E}}{\omega}\right) = \frac{q}{\omega}\left[(\underline{v_d}\cdot\underline{E})\underline{k} - (\underline{v_d}\cdot\underline{k})\underline{E}\right]$$

$(\underline{v_d}\cdot\underline{k}) \to 0$

$$= \underbrace{q\underline{E}}_{\underline{F_e}}\cdot\underline{v_d}\cdot\frac{\underline{k}}{\omega}\cdot\underline{e_k} = \frac{\underline{F_e}\cdot\underline{v_d}}{c}\cdot\underline{e_k}$$

$\underline{F}\cdot\underline{v}$: power transmitted to e^- ! Joule heat!

$$\frac{\partial E_{Joule}}{\partial t}$$



$$\frac{\partial E_{EM}}{\partial t} = -\oint \underline{E}\times\underline{H}\,d\underline{s} - \frac{\partial E_{joule}}{\partial t}\Big/\langle\ \rangle$$

$$0 = -\Delta A\langle\underline{S}\rangle - \left\langle\frac{\partial E_{joule}}{\partial t}\right\rangle$$

$$\underline{F_B} = \frac{\Delta A\cdot\langle\underline{S}\rangle}{c} \text{ for one } e^- \quad / \quad \underline{F_{BN}} = \frac{\Delta A\langle\underline{S}\rangle}{c}$$

the pressure :

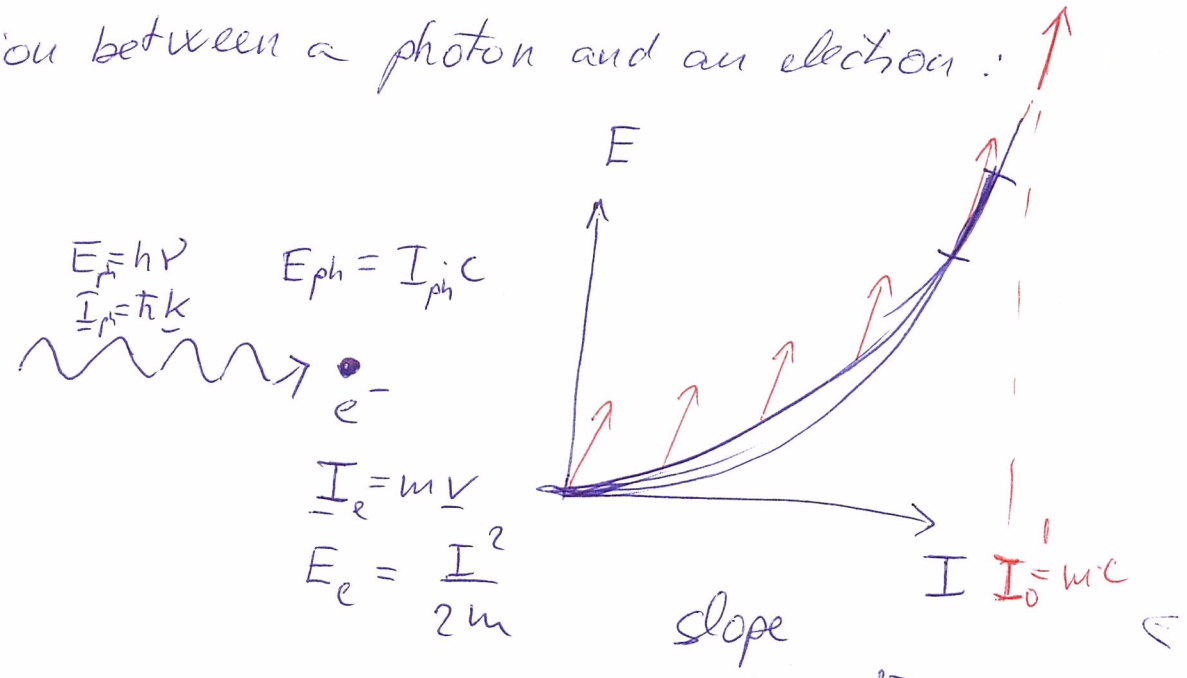
$$p = \frac{\Sigma F}{\Delta A} = \frac{|\langle\underline{S}\rangle|}{c}$$ (if all energy is absorbed !)

$p = \frac{2\langle S\rangle}{c}$ complete reflection !

COMO IKARUS successful solar sailing mission

Challenge: Is it possible To

collision between a photon and an electron:



$$\frac{dE_{ph}}{dI} = c$$

$$\frac{dE_e}{dI} = \frac{I}{m} = v$$

$\frac{dE}{}$

We need a second massy particle!

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for a relativistic e^-

$$E_e = c\sqrt{m_0^2 c^2 + I_e^2}$$

$$\frac{dE}{dI} = \frac{c}{2\sqrt{m_0^2 c^2 + I_e^2}} \cdot 2I$$

$$\frac{dE}{dI} = \frac{c\,I}{\sqrt{m_0^2 c^2 + I^2}}$$

$v \ll c$ $\quad I \ll m_0 c$ $\quad \frac{dE}{dI} = \frac{c\,I}{m_0 c} = 1$

$v \approx c$ $\quad \frac{dE}{dI} = \frac{c\,m_0 v}{\sqrt{m_0^2 c^2 + 2}} = \frac{c m_0 v}{m_0 v \sqrt{2}} = \frac{c}{\sqrt{2}}$ does not work

Challenge? 3 body interaction

Newton II: ↓ momentum!

$$m \cdot \underline{a} = \frac{d\underline{I}}{dt} = \underline{\Sigma F} = \delta A \cdot \frac{\langle \underline{S} \rangle}{c} = \frac{\partial E_{joule}}{\partial t} \cdot \frac{1}{c} \cdot \underline{e}_k$$

$$\frac{dI}{dt} = \frac{dE_{EM}}{dt} \cdot \frac{1}{c} \Rightarrow I = \frac{E_{EM}}{c}$$

the momentum of the EMW ∼ to its energy!

$$E_{EM} = I_{EM} \cdot c$$

I grows in a material if $v_{ph} < c$

(For a ~~mechanical~~ massy body:

$$E = \frac{1}{2} m v^2 = \frac{1}{2m} (mv)^2 = \frac{I^2}{2m}$$

you are going to learn in the theory of spec. relativity:

$$E = c\sqrt{m_0^2 c^2 + I^2} = cI$$

if $m_0 = 0$ (for $v \ll c$ $E = c\sqrt{m_0^2 c^2 + m_0^2 v^2} =$

$$= m_0 c^2 \sqrt{1 + \frac{v^2}{c^2}} \cong m_0^2 c^2 + \frac{1}{2} m_0 c^2 \frac{v^2}{c^2}$$

$$= m_0^2 c^2 + \frac{1}{2} m_0 v^2 \;)$$

To observe light pressure you have to experiment in a very good vacuum!

~~Lebedev~~ Pyotr Lebedev measured it in a high vacuum!

Crookes radiometer  turning wrong dir.!

$p_{atm} = 10^5$ Pa

$p_{exp} = 10^{-6}$ p.

Spectrum of EMW

$$c = \frac{\omega}{k} = \frac{\lambda}{2\pi} \cdot \omega = \lambda \cdot f \quad \text{in vacuum!}$$

(f)

$$\lambda_0 = \frac{c}{f}$$

$$\lambda_n = \frac{c}{n f}$$

Long waves: propagate far, reflected by ionosphere